

Mechanisms of Obesity and Obesity Management

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Block: Endocrine/ Urology/Reproductive

Thread: Clinical Medicine/Physiology

Conflicts of Interest or disclosures: none

Learning Objectives

1. Explain the regulation and pathophysiology of appetite and energy balance that contribute to obesity.
2. Formulate a therapeutic management plan to treat obesity.
3. Describe the clinical use, contraindications, mechanism of action, side effects and efficacy of each class of drugs used to treat obesity.
4. Identify which patients may benefit from surgical options to treat obesity.

Study Questions

1. After a long day of attending lectures and studying, you and your friends go out to eat dinner. In general terms, what would you expect of the relative serum concentrations of these hormones prior to, and immediately following, the meal: ghrelin, GLP-1, GIP, leptin and insulin? (ie, Which ones are high before meal, low before the meal, and vice versa after the meal)?
2. What is hedonic hunger and what environmental stimuli may be triggering your hedonic hunger drive as you go about your day?
3. What evidence presented in this talk supports the idea that our bodies maintain defended fat masses?
4. What are the non-pharmacologic, non-surgical treatment options available for patients living with overweight or obesity? When would you consider using each?
5. Which class of anti-obesity medication is the most effective available?
6. When should you consider referral to bariatric surgery?

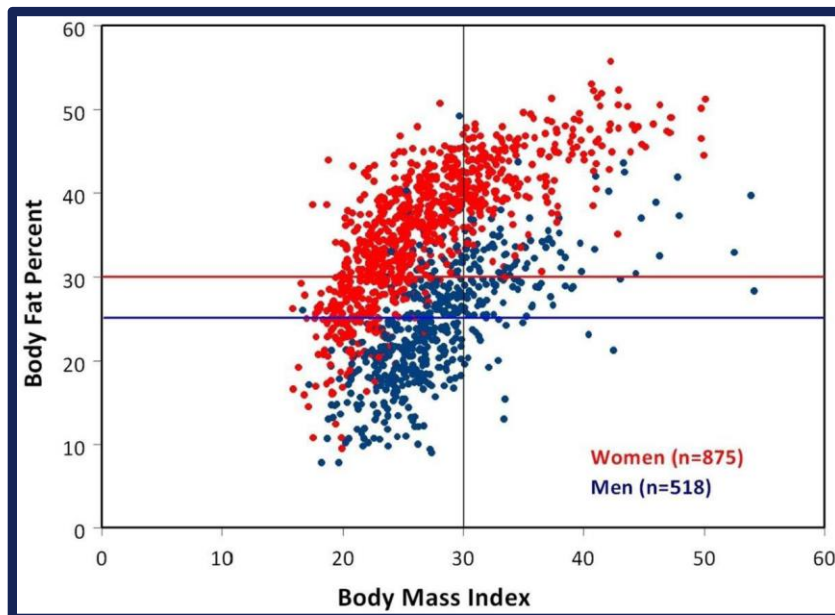
A. Outline

1. Define and classify obesity
2. Describe the physiology of hunger and pathophysiology of obesity
3. Discuss patient diagnosis and evaluation
4. Discuss treatment options (behavioral, pharmacologic, surgical)

B. Definition of Obesity

1. It is a chronic disease of excess adipose deposition caused by disruption of the normal physiology of hunger and energy balance often resulting in complications which may be debilitating or life-threatening.

C. Body Mass Index (BMI) as an indicator of body fat percentage



BMI is easy to measure and remains in wide use but is not reliable as an indicator of body fat percentage (body composition) nor health.

D. Classification of Obesity

1. Adults

Classification	BMI (kg/m ²)	
	International classification	Asian population*
Underweight	<18.5	
Healthy weight	≥18.5 and <25	≥18 and <23
Overweight	≥25 and <30	≥23 and <25
Obesity	≥30	>25
Obesity class I	≥30 and <35	
Obesity class II	≥35 and <40	
Obesity class III	≥40	

- a. $BMI = \text{Weight (kg)} / [\text{Height (m)}]^2$
- b. Asian populations: Obesity-related comorbidities are more prevalent and occur at lower BMI in the Asian population. Treatment should be considered at lower BMI.
 - i. <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC8865785/>

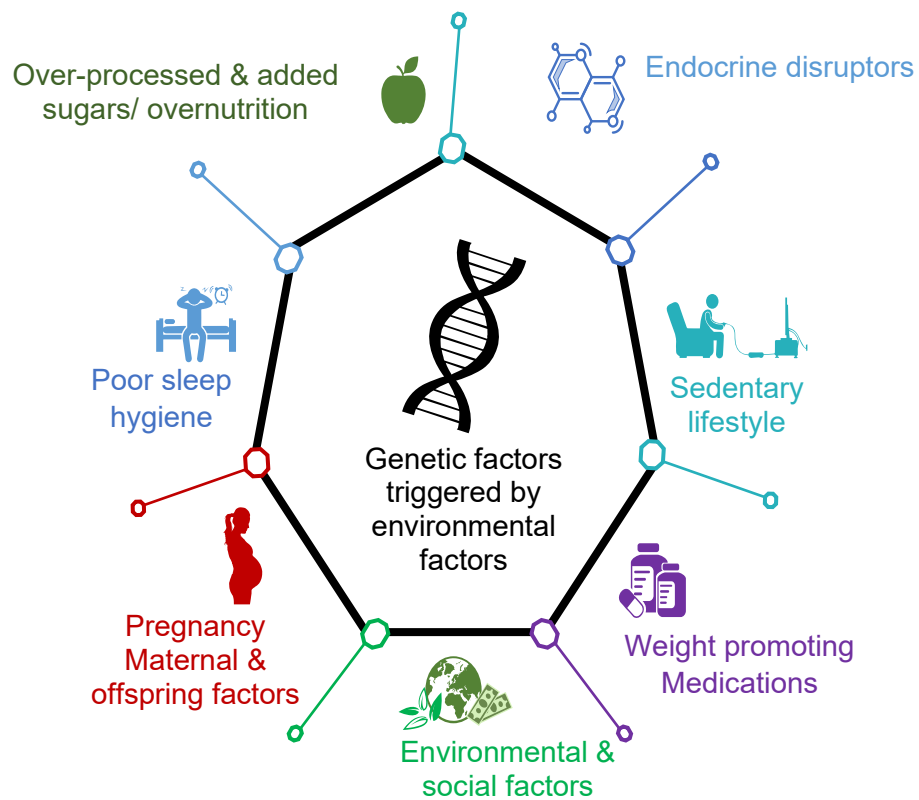
2. Children

- a. In children ages 2 to 19, BMI is assessed by age- and sex-specific percentiles:
 - i. Underweight: BMI < 5th percentile
 - ii. Normal weight: BMI > 5th and < 85th percentile
 - iii. Overweight: BMI > 85th and < 95th percentile
 - iv. Obesity: BMI > 95th percentile
 - v. In children <2, CDC uses a modified version of WHO criteria

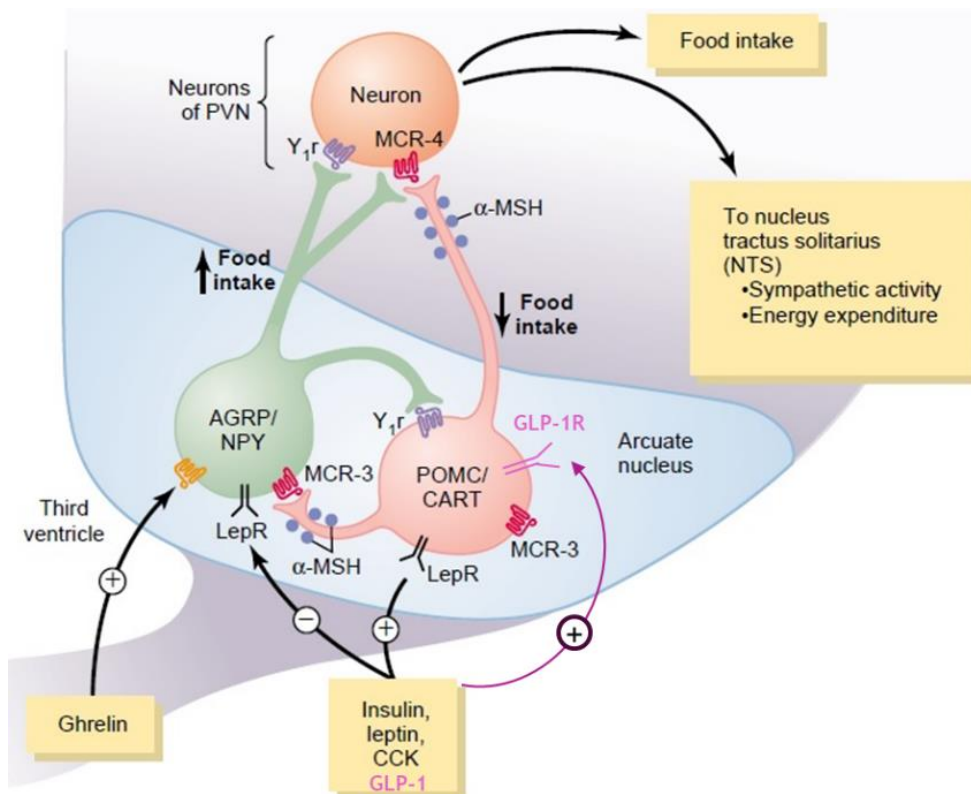
3. Obesity Prevalence in The United States

- a. According to NCHS Data Query System
 - i. Approximately 40% of adults in the US meet criteria for obesity.
 - ii. Approximately 20% of children in the US meet criteria for obesity.
 - iii. <http://www.cdc.gov/nchs/dqs/index.html>

E. Physiology of Hunger and Satiety: The Current Understanding



F. Physiology of Hunger and Satiety: Central Biochemical Pathways



1. Diagram: Arcuate Nucleus of the Hypothalamus
 - a. Orexigenic pathway stimulation (left) triggers increased appetite, increased feeding, and decreased energy expenditure
 - i. Hunger Hormones: **ghrelin**
 - b. Anorexigenic pathway stimulation (right) triggers decreased appetite, decreased feeding and increased energy expenditure
 - ii. Satiety Hormones: amylin, cholecystokinin, **GIP**, **GLP-1**, **insulin**, **leptin**, oxyntomodulin, peptide YY

G. Physiology of Hunger

Three Hunger Mechanisms: homeostatic, hedonic and microbiota-driven

1. Homeostatic hunger – dominant when food is scarce
 - a. Generally, does not result in excessive calorie intake
 - b. Primary mechanism: ghrelin secretion by the pancreas, small intestine and stomach leads to activation of the orexigenic pathway
 - i. Secretion of ghrelin is triggered by low serum concentrations of amino acids, fatty acids & glucose

- c. Secondary mechanism: gastric mechanoreceptor quiescence triggers vagal-mediated orexigenic neuronal stimulation
- 2. Hedonic hunger – dominant when food is abundant
 - a. Characterized by “de-coupling” of appetite and acute caloric need
 - b. Supersedes homeostatic mechanisms resulting in calorie intake for reasons other than physiologic hunger
 - i. There is preferential intake of energy-dense, high-fat, and high-sugar foods
 - c. Triggered by many stimuli:
 - i. Emotions lead to reward-seeking behaviors (including food)
 - ii. Anticipation of feeding (olfactory, visual and auditory stimuli)
 - iii. Taste preferences, cultural beliefs and religious practices
 - iv. Socioeconomic factors: unhealthful food often less costly, urban and rural food deserts, breastfeeding practices
- 3. Microbiota-driven hunger
 - a. Data are observational, animal-based, currently of limited utility
 - b. Gut microbes capable of influencing the release of hunger and satiety hormones have been identified
 - c. Some gut microbes have been shown to produce peptidomimetic proteins (that mimic structure and function of endogenous hormones)

H. Metabolic adaptations to Obesity, Set Point Theory

- 1. Research suggests the body maintains a defended fat mass.
- 2. There are persistent metabolic adaptations to obesity (after weight loss).
- 3. We try to lower a patient’s fat mass set point to achieve long-term weight loss.
 - a. This is achieved by modifying the patients’ hedonic and microbiota-related habits with lifestyle modifications, medications and/or surgery.

I. Evaluation of patients with overweight and obesity

- 1. Take a good history
- 2. Perform a good exam
- 3. Review objective data
 - a. Vital signs: make sure the right cuff size is used
 - b. Markers of metabolic health: A1c, lipids, CMP (renal function, liver enzymes), thyroid function testing, sleep study reports, liver imaging, etc.
 - c. Optional: genetic testing if there is suspicion for monogenic or syndromic obesity
- 4. Assessment should include: severity of illness, presence of complications, and response to treatment(s)

J. Treatment of patients with overweight and obesity

1. Treatment requires an inter-disciplinary and multimodal approach including:
 - a. Lifestyle modifications – all team members
 - b. Medical nutrition therapy – registered dietitian
 - c. Behavioral therapy – behavioral therapy experts
 - d. Anti-obesity medications (AOMs) – obesity medicine providers
 - e. Bariatric surgery – bariatrics providers

1. Anti-obesity Medications

- a. Always used in conjunction with lifestyle modifications
- b. Criteria for AOM Use:
 - i. $<5\%$ total body weight loss after 3 months of lifestyle modifications AND
 - ii. $\text{BMI} \geq 27 \text{ kg/m}^2$ with at least 1 life-threatening complication of obesity OR
 - iii. $\text{BMI} \geq 30 \text{ kg/m}^2$
- c. General approach:
 - i. Start at low dose, up-titrate to minimum dose necessary to achieve goal(s)
 - ii. Typically prescribed on a long-term basis (which is off-label for some)
- d. No AOM is currently considered safe in pregnancy or breastfeeding.

2. Current FDA-approved pharmacotherapy: Overview

- a. Orlistat
 - i. Mechanism of Action: lipase inhibitor resulting in decreased capacity to breakdown and absorb lipids within the GI tract
 - ii. Contraindications: cholestasis, chronic malabsorption syndrome, hypersensitivity, pregnancy
 - iii. Common side effects: abdominal discomfort, abdominal pain, fecal urgency, anal leakage, steatorrhea
 - iv. *Note: Alli is a lower-dose over-the-counter formulation, Xenical is a higher-dose prescription formulation*
- b. Phentermine, Diethylpropion
 - i. Mechanism of Action: stimulate norepinephrine release resulting in appetite suppression (through an unknown mechanism)
 - ii. Contraindications: cardiovascular disease, glaucoma, hyperthyroidism, uncontrolled hypertension, hypersensitivity, pregnancy

- iii. Common side effects: agitation, anxiety, constipation, insomnia, tachycardia, xerostomia
 - iv. Rare, but serious, side effects: arrhythmia, acute glaucoma, acute coronary syndrome
 - v. *Note: studies regarding side effects are conflicting; seems to have low abuse potential*
- c. Naltrexone-bupropion ER
- i. Mechanism of Action: opioid antagonist, dopamine/norepinephrine reuptake inhibitor resulting in appetite suppression (through unknown mechanisms)
 - ii. Contraindications: concomitant opioid use (naltrexone may precipitate withdrawal), seizure disorder (bupropion lowers seizure threshold), uncontrolled hypertension, hypersensitivity, pregnancy
 - iii. Common side effects: constipation, nausea, vomiting, dizziness, insomnia, headache
 - iv. *Note: naltrexone is indicated separately for alcohol and opioid use disorders; bupropion is indicated separately for smoking cessation, major depression, and seasonal affective disorder*
- d. Phentermine-topiramate ER
- i. Mechanism of Action: stimulate norepinephrine release and modulates CNS GABA receptors resulting in appetite suppression (through unknown mechanisms)
 - ii. Contraindications: cardiovascular disease, glaucoma, hyperthyroidism, uncontrolled HTN, MAOI use within 14 days, hypersensitivity, pregnancy
 - iii. Common side effects: agitation, insomnia, xerostomia, constipation, altered taste, paresthesia, dizziness
 - iv. *Note: topiramate is also indicated as an antiepileptic, mood stabilizer, and for migraine prevention*
- e. Liraglutide, Semaglutide, Orforglipron
- i. Mechanisms of Action: GLP-1 receptor agonists indirectly stimulate POMC neurons in the arcuate nucleus of the hypothalamus promoting satiety, inhibit the limbic (reward) system, and directly stimulate GLP1Rs throughout GI tract to slow gastric emptying (among other actions)
 - ii. Refer to DM2 pharmacology lecture for contraindications and side effects
 - iii. *Note: Saxenda is a once-daily injection*

- iv. Note: Wegovy is a once-weekly injection and is also FDA-approved for cardiovascular risk reduction in people with a history of myocardial infarction, history of stroke or diagnosis of peripheral artery disease
- v. Wegovy is also FDA-approved for treatment of Metabolic dysfunction-Associated Steatohepatitis (MASH) with moderate or advanced fibrosis
- f. Tirzepatide
 - i. Mechanism of Action: GLP-1/GIP receptor agonists indirectly stimulate POMC neurons in the arcuate nucleus of the hypothalamus promoting satiety, inhibit the limbic (reward) system, and directly stimulate GLP1Rs throughout GI tract to slow gastric emptying (among other actions)
 - ii. Refer to DM2 pharmacology lecture for contraindications and side effects
 - iii. Note: Zepbound is a once-weekly injection also FDA-approved for the treatment of moderate or severe obstructive sleep apnea
- g. Setmelanotide
 - i. Indicated for severe obesity caused by certain genetic disorders in patients ≥ 2 years old
 - 1. Bardet-Biedl Syndrome
 - 2. POMC Receptor Mutation
 - 3. PCSK1 Mutation
 - 4. Leptin Receptor Mutation
 - ii. Mechanism of Action: Melanocortin-4 receptor (MC4R) agonist resulting in decreased feeding behavior and increased energy expenditure
 - iii. Contraindications: history of hypersensitivity reaction
 - iv. Most common side effects: hyperpigmentation, increased libido, hypersensitivity
 - v. Note: Imcivree is a once-daily injection

Anti-obesity Medication	Efficacy (%TBWL)	Relative Cost
Orlistat	3	\$
Naltrexone-Bupropion	4	\$\$
Phentermine	7	\$
Liraglutide	8	\$\$
Phentermine-Topiramate	8	\$\$
Orforglipron	12	\$\$\$
Setmelanotide	13	\$\$\$\$\$
Semaglutide oral	14	\$\$\$
Semaglutide injection	16	\$\$\$
Tirzepatide	20	\$\$\$

3. Bariatric Surgery

- a. Indications: $\text{BMI} \geq 35 \text{ kg/m}^2$ or $\text{BMI} \geq 30 \text{ kg/m}^2$ with comorbidity
- b. Post surgery, lifelong vitamin and supplement use and monitoring required
- c. Most Common Bariatric Surgeries
 - i. Sleeve Gastrectomy
 - ii. Gastric Bypass
 - iii. Duodenal Switch
- d. After surgery, lifelong vitamin administration and monitoring recommended
- e. Bariatric surgery has been shown to decrease mortality

Answers to Study Questions

1. After a long day of attending lectures and studying, you and your friends go out to eat dinner. In general terms, what would you expect of the relative serum concentrations of these hormones prior to, and immediately following, the meal: ghrelin, GLP-1, GIP, leptin and insulin? (ie, Which ones are high before the meal, low before the meal, and vice versa after the meal)?

Answer: Prior to the meal, we would expect your serum concentration of ghrelin to be elevated and your serum concentration of GIP, GLP-1, insulin and leptin to be low. After the meal, we would expect your serum concentration of ghrelin to be low and your serum concentration of GIP, GLP-1, insulin and leptin to be elevated.

2. What is hedonic hunger and what environmental stimuli may be triggering your hedonic hunger drive as you go about your day?

Answer: Hedonic hunger is the drive to eat in the absence of acute caloric need. This drive may be triggered by emotions (positive or negative), thinking about food (advertisements), seeing food (advertisements), smelling food (walking by a restaurant), cultural beliefs, religious practices, taste preferences

3. What evidence presented in this talk supports the idea that our bodies maintain defended fat masses?

Answer: The data presented by Sumithran, et al, in the NEJM 2011 showing that baseline concentrations of ghrelin and the satiety hormones never return to “normal” during, and after, someone experiences weight loss.

4. What are the non-pharmacologic, non-surgical treatment options available for patients living with overweight or obesity? When would you consider using each?

Answer: sleep optimization, stress reduction and management, nutrition optimization and physical activity. These should be considered with EVERY patient.

5. Which class of anti-obesity medication is the most effective available?

Answer: The GLP-1/GIP dual agonist class (ie tirzepatide).

6. When should you consider referral to bariatric surgery?

Answer: For any patient with a BMI ≥ 30 kg/m² with metabolic complications of obesity (diabetes, hyperlipidemia, hypertension, etc.) and any patient with BMI ≥ 35 kg/m² and no compelling reason to avoid surgery.